LEXIGRID

Upgrade, Backup and Recovery

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Procedures, examples, verification and operational limits. Complete commissioning on your deployment before serving customer traffic. Sample values are fictional; original keys and data must be preserved.

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Upgrade, backup and recovery

Purpose and prerequisites

This distribution initializes an empty database. Keep the original installer and backup for existing deployments. Use this distribution to restore backups created from this schema.

Restore data, tenant boundaries and runtime permissions together. A database dump alone is not a complete recoverable deployment. Retain the compatible code release, VERSION, schema revision, original evidence/MFA/mail/audit/anchor keys, identity trust configuration, logging configuration and required external-provider access. Store secrets separately under controlled recovery access.

Verify the restored schema against the application before starting the service. Investigate duplicate reminder schedules without deleting records automatically.

PostgreSQL backup

1. Use the approved database backup role/owner connection and a private backup directory. The script expects a libpq PostgreSQL URL, not the SQLAlchemy +psycopg driver suffix.

2. Run scripts/backup_postgres.sh using approved secret injection. Avoid command-line password literals and shell history containing credentials.

3. Keep the generated .dump and adjacent .sha256 manifest together. The backup contains sensitive tenant data; protect/encrypt it before transfer to independent storage.

4. Verify protected file permissions and the manifest. A checksum is integrity evidence, not encryption or proof that a restore works.

5. Restore a copy into an isolated database and verify application access before calling the backup recoverable.

The backup script uses unique staged names to avoid same-second collisions. It must not overwrite an earlier archive as a routine way to make a backup succeed.

PostgreSQL isolated restore

1. Provision an isolated target schema owner and restricted lexigrid_runtime login. Prevent customer traffic and workers from reaching it.

2. Retain the supplied archive and checksum sidecar. Set RESTORE_CONFIRM=RESTORE_LEXIGRID only for the intended recovery target.

3. Set DATABASE_URL to the target owner's libpq URL and invoke scripts/restore_postgres.sh ARCHIVE_PATH. This is a data-replacing operation; do not target the active database casually.

4. The script hashes the supplied archive even if a legacy manifest refers to an old absolute path. Tampering must be rejected before restore.

5. Archive restoration, runtime grants, forced-RLS and history privilege checks execute in one transaction. SQL failure rolls back the restore.

6. Restore an archive with the schema for this distribution. Keep a copy of the compatible installer with each protected backup.

7. Verify the restored schema with deployment diagnostics. Configure the API with runtime credentials and original keys.

Expected result: preserved counts and selected content fingerprints, no foreign-tenant visibility, denied runtime history mutation, denied protected outbox/delivery deletion and a healthy compatible API. A restore returning success is not the final traffic-switch approval.

SQLite local recovery

SQLite is development/evaluation mode. Use SQLite's online backup API for a consistent copy; copying an active database file alone can miss WAL state. Create the destination privately before writing, run PRAGMA integrity_check and foreign_key_check, and retain an unchanged original snapshot.

Restore to a separate path. Run Alembic against that copy, compare all original table counts/content fingerprints and cold-start updated code with outbound workers disabled. A verified copy is the precondition for an approved installed upgrade, not permission to experiment on the only original database.

For a quiet installed upgrade, stop/verify the native instance, finish outstanding work, take a final snapshot, validate the final restored copy, migrate the intended database and check all original records before restart. Preserve the bind address and environment keys. If validation fails, keep traffic stopped and use the verified compatible recovery path.

Application and key verification

Authenticate a designated test account in the intended tenant and verify one existing record, an evidence download, configuration decryption, an audit read and a foreign-tenant denial. Test each required IdP and MFA policy independently. Original local evidence may require the legacy evidence key even after new writes select Vault or Conjur. A restored mail configuration needs the original shared LEXIGRID_MAIL_CONFIG_KEY.

Do not generate replacement keys to solve a read error. Provider 503 may mean lost access, credentials, CA trust or missing historical key availability; investigate before declaring corruption. Failed authentication/hash checks remain integrity errors. Key recovery/rotation must be exercised with the actual provider.

Worker restart and duplicate semantics

Re-enable only the designated workers/scheduler after the restored database is verified. Inspect queued/failed/sending/retry states and leases before manual recovery. A process may have sent a message before recording success. Keep stable notification IDs and gateway deduplication; do not reset attempt counters to suppress evidence of retries.

Acceptance record and rollback trigger

Record release/schema, backup manifest identity, protected recovery location, counts/fingerprints, runtime role, tenant-negative tests, key/evidence reads, worker decisions and traffic-switch result. Keep credentials out of that record. Missing records, changed existing fingerprints, a privileged API role, broken RLS, unavailable original keys or an unhealthy cold start are rollback/hold conditions.

Complete a recovery exercise against an isolated target. Verify archive checksums, expected row counts, restricted runtime grants, tenant reads and original key recovery before approving production recovery. Test HA and failover separately against the intended infrastructure.

Dedicated PostgreSQL identities

Database identities must be explicit and separate from an operator's operating-system login. Native installation defaults to `lexigrid_owner` for privileged setup, and the application uses restricted `lexigrid_runtime` credentials after provisioning. A DBA may choose a dedicated maintenance account such as `lexiadmin` using `--pg-user lexiadmin`; create it under the deployment's approved database administration procedure. Migration-created roles use fixed product names, never the current OS username. The installer does not automatically create a personal or privileged account named after the person running it. SQLite uses a database file and has no database login.

For a supplied PostgreSQL URL, put the intended account explicitly before `@`, for example `postgresql+psycopg://lexigrid_runtime@db.example.invalid/lexigrid` (supply credentials through the approved configuration or secret mechanism). URLs such as `postgresql:///lexigrid` and URLs with `user` / `username` query overrides are rejected before a connection is attempted. Installation, application startup, maintenance, runtime provisioning, diagnostics and the backup script apply the same check. The operating-system username, PGUSER and database-service defaults cannot supply an omitted identity.

Before upgrading an existing PostgreSQL deployment with an implicit username URL, have the DBA supply the approved dedicated account explicitly. Do not rename roles, reset passwords, replace encryption keys or recreate databases simply to satisfy this setting. Existing explicit connections, original credentials and data are retained. Audit records may still identify the actual OS operator separately from the database role.